

1. Introduction

Fractal characteristics are ubiquitous in nature, ranging from the branching patterns of vegetation to the large-scale structure of the universe[1,2]. In the geological domain, subterranean rock formations and fracture networks exhibit pronounced self-similarity over a wide range of scales[3,4]. Both naturally occurring fractures and those induced by hydraulic fracturing possess these fractal features, which are critical in governing the flow and transport properties within porous media[5]. Understanding the transport mechanisms in such multiscale systems is fundamental to applications in enhanced oil recovery, groundwater remediation, and carbon sequestration.

Among the key parameters characterizing transport, tortuosity (τ) plays a pivotal role. It serves as a bridge connecting the microscopic pore structure to macroscopic transport properties, such as permeability and electrical conductivity[6,7]. In classical theories, such as the Kozeny-Carman equation, tortuosity is defined as the ratio of the actual fluid path length to the straight-line distance across the medium[8,9]. Accurately quantifying tortuosity is essential for predicting flow resistance and diffusion rates[10]. Traditionally, tortuosity has been analyzed using simplified geometric models (e.g., capillary bundles) or numerical simulations based on Euclidean geometries[11–13]. While these approaches have established robust baselines for homogeneous media, they often struggle to capture the extreme heterogeneity and connectivity constraints inherent in disordered geological materials[14,15]

In recent years, fractal theory has provided a powerful tool to describe these complex, non-Euclidean structures. Numerous studies have derived analytical models linking fractal dimensions (D_f) to transport parameters[16,17]. For example, models based on deterministic fractals, such as the Sierpiński carpet, have highlighted the effects of fractal dimensions and porosity on tortuosity, providing deeper insights into flow paths[18]. Furthermore, researchers have utilized fractal geometry to refine the analysis of seepage and imbibition, demonstrating that flow resistance is significantly influenced by the scaling laws of pore size distributions[19].

Despite these advancements, a fundamental gap remains: most existing studies operate within a "Euclidean framework with fractal corrections." Traditional models typically treat fractal dimensions merely as fixed geometric parameters entered into classical conservation equations (e.g., standard Fickian diffusion or Darcy's law). They implicitly assume that the metric of the space remains Euclidean, where distance scales linearly[20]. However, transport in *intrinsic fractal spaces* is fundamentally different. As established in the physics of disordered media, diffusion on fractals is often anomalous (sub-diffusive), governed by the spectral dimension rather than the Euclidean dimension[21]. In such spaces, the concept of a constant tortuosity becomes questionable. The fluid path is not merely longer; it is structurally constrained by topological features—such as dead ends, bottlenecks, and back-turns—that exist at all scales[22]. Consequently, classical models fail to capture the scale-dependent nature of transport pathways in fully fractal environments.

This study aims to bridge the gap between classical tortuosity studies and the novel

implications of fractal geometry. Unlike previous works that use deterministic fractals, we construct a random porous media model explicitly within a fractal space framework, utilizing an improved Random Sierpiński Carpet to mimic the stochastic nature of natural rocks, and evaluate it using a restricted finite random-walk framework[23]. By exploring transport dynamics within this non-Euclidean topology, this work seeks to quantify how the intrinsic properties of fractal space—specifically the coupling between porosity ϕ and fractal dimension (D_f)—alter transport laws, thereby providing a rigorous physical basis for describing flow in complex reservoirs.

2. The random porous media in fractal space

The random porous medium discussed in this paper is represented by an $n \times n$ matrix, where each element consists of either 0 or 1, with 1 indicating solid material and 0 representing the pore spaces through which fluid can flow. In constructing a random porous medium with a given porosity using the Monte Carlo method, a random number is first generated. For each matrix element, a uniform random number is generated and compared against the specified porosity. If the random number exceeds the specified porosity, the corresponding element is assigned a value of 1; otherwise, it is assigned a value of 0. Iterating this procedure over a sufficiently large $n \times n$ domain yields a baseline random porous medium with the desired macroscopic porosity.

The Sierpiński carpet is an ideal fractal structure in two dimensions. The standard construction process of the Sierpiński carpet involves dividing a square into 9 equal smaller squares, removing the central square, and retaining the remaining 8 squares. This process is then recursively applied to the 8 remaining smaller squares, continuing infinitely to form the Sierpiński carpet. To generate a random porous medium in a fractal space, this paper utilizes a finite number of iterations to create the Sierpiński carpet as the fractal space. In this fractal space, the random porous medium with a given porosity can be obtained using the previously described method.

To better study the tortuosity of random porous media in fractal spaces, we replace the standard Sierpiński carpet with a random Sierpiński carpet. There are three main modifications: First, the square is not uniformly divided into 9 equal smaller squares, but also into squares of sizes 4×4 , 5×5 , and 6×6 , among others. Second, the number of squares removed in each iteration is fixed but not necessarily equal to 1. Third, the positions of the removed squares are randomized during each iteration. Through this approach, we can obtain fractal spaces with different fractal dimensions, and the calculation formula for the fractal dimension D_f is as follows[24]:

$$D_f = \frac{\log(m^2 - k)}{\log m} \quad (1)$$

where m represents the number of subdivisions along each side of the square during each iteration, and k denotes the number of squares removed at each step.

As illustrated in Fig.1 (a), when $m = 5$ and $k = 4$, the fractal dimension of the resulting random Sierpiński carpet generated through iteration is 1.892. The Fig.1 (b) denotes the random porous media in fractal space with porosity $\phi = 0.5$ and fractal

dimension 1.861. The Fig.1 (b) appears visually less contrasted because additional randomly distributed pores are introduced within the solid regions of the fractal support. This superposition of stochastic heterogeneity onto the underlying fractal geometry leads to fine-scale mixing of black and white pixels, producing a macroscopically gray appearance.

(a) (b)

Figure 1. (a) Random Sierpiński carpet constructed by finite iterations, representing the fractal support space ($m = 5$, $k = 4$, fractal dimension $D_f = 1.892$); (b) Random porous medium embedded in the fractal space with porosity $\varphi = 0.5$ and fractal dimension $D_f = 1.861$.

Solid regions are shown in white and pores in black.

3. Numerical algorithms for tortuosity

According to the definition of tortuosity, we can calculate the flow field using fluid dynamics methods and obtain the streamlines by analyzing the pathlines in steady-state flow, from which the streamline length can be determined and tortuosity calculated. However, this approach is computationally intensive and presents modeling challenges, especially for random porous media. The presence of a large number of small, randomly distributed solid particles in the medium adds complexity to the calculations. To address the need for efficient tortuosity calculations in large-scale fractal-space random porous media, this paper employs an efficient finite random walk method[23].

In this method, particles are released from the pore at the edge of the random porous medium flow domain (typically chosen as the left edge), where each pore is represented as "0." The particles attempt to move toward the opposite edge. If the direction of movement is not "0," the particle randomly chooses a direction perpendicular to its current path and continues to attempt to move toward the opposite edge. If the particle encounters a dead-end or flow boundary, it backtracks to the most recent random choice and selects another direction. This process continues until the particle either reaches the opposite edge or returns to the starting point. The path of particles successfully reaching the opposite edge is recorded as a streamline. By analyzing the lengths of all streamlines, tortuosity can be calculated.

The tortuosity values for random porous media with different porosities in Euclidean space, calculated using this method, show that in the two-dimensional case, due to the percolation threshold, the theoretical and computed values align closely only when the porosity exceeds 0.7. In the three-dimensional case, the better connectivity leads to an even closer match between the theoretical and computed values. This validates the effectiveness of the finite random walk method. Therefore, this method is employed in this paper to calculate the tortuosity of random porous media in fractal spaces.

4. Results and analysis of tortuosity in random porous media in fractal spaces

In order to thoroughly analyze the tortuosity of random porous media in fractal

spaces, the finite random walk method was employed to study a series of fractal-space random porous media with different porosities and fractal dimensions. When calculating the tortuosity of random porous media in a fractal space with a specific porosity and fractal dimension, the inherent randomness of the porous structure must be considered. Since the subject of the numerical study cannot be infinite in size, and to reduce the bias introduced by randomness, the tortuosity data for each case were averaged over multiple sets of data with the same fractal dimension and porosity. Convergence tests indicate that averaging over 200 independent realizations for each (φ, D_f) combination yields an optimal balance between computational efficiency and statistical stability. Therefore, all the tortuosity data presented are the average values of 200 data sets. The comprehensive dataset of the ensemble-averaged tortuosity and corresponding standard deviations for all (φ, D_f) configurations is provided in Supplementary Table 1.

4.1 Statistical characteristics of tortuosity in fractal porous space

Table 1 summarizes the statistical properties of tortuosity obtained from repeated realizations for each combination of porosity and fractal dimension. For each fractal configuration, multiple independent realizations were generated, and the mean tortuosity and corresponding standard deviation were calculated to reduce stochastic fluctuations induced by random walks.

As shown in Fig. 2, the tortuosity generally decreases with increasing porosity, indicating that higher pore availability leads to more direct transport pathways. However, for a given porosity, noticeable differences in tortuosity are observed among different fractal dimensions, suggesting that pore connectivity and geometric complexity play a non-negligible role in controlling transport efficiency.

Fig.2 Tortuosity vs Porosity (grouped by D_f)

4.2 Dependence on fractal dimension and porosity

Fig.3 (a) presents the variation of tortuosity τ with fractal dimension D_f at three representative porosities ($\varphi = 0.72, 0.83$, and 0.93). The results demonstrate that the influence of D_f on τ is not strictly monotonic and depends strongly on porosity. At low porosity ($\varphi = 0.72$), increasing D_f generally increases tortuosity, indicating that fractal embedding enhances pathway complexity when the pore space is more constrained. In contrast, at higher porosity ($\varphi = 0.93$), the dependence of τ on D_f becomes much weaker and can even reverse, implying that porosity-driven connectivity dominates and mitigates the geometric constraints introduced by the fractal space. To provide a global overview of this coupled dependence, Fig.3 (b) shows a two-dimensional map of $\tau(D_f, \varphi)$ with labeled contour lines. The map corroborates the trend observed in Fig.3(a) by revealing a structured (D_f, φ) dependence rather than a purely monotonic behavior with respect to D_f . Specifically, the contours indicate that the sensitivity of τ to fractal geometry is strongest at relatively low φ , whereas at high φ the tortuosity varies predominantly with porosity and exhibits reduced dependence

on D_f .

Fig. 3 (a) Tortuosity τ versus fractal dimension D_f at representative porosities ($\phi = 0.72, 0.83$, and 0.93). Symbols denote numerical results (error bars indicate 95% confidence intervals of the mean when available). Lines are drawn to guide the eye and broken at missing D_f intervals to avoid misleading connections.

Fig. 3 (b) Heat map of $\tau(D_f, \phi)$ over the (D_f, ϕ) parameter space. Labeled contour lines highlight regimes of similar tortuosity and provide a global overview of the coupled dependence on porosity and fractal dimension.

To quantitatively describe the interaction between porosity and fractal dimension, a logarithmic regression model including an interaction term was employed:

$$\ln \tau = 1.3945 + 3.1399 \ln \phi - 0.6601 D_f - 2.0723 D_f \ln \phi \quad (2)$$

where τ is the tortuosity, ϕ is the porosity, and D_f is the fractal dimension. The model yields $R^2 = 0.81$, indicating good agreement with the numerical results and capturing the observed porosity-dependent reversal in the D_f effect. Notably, the slope with respect to D_f is

$$\frac{\partial \ln \tau}{\partial D_f} = -0.6601 - 2.0723 \ln \phi \quad (3)$$

which can be positive at low ϕ (since $\ln \phi < 0$) but becomes negative at high ϕ , consistent with Fig.3 (a).

This expression can be rewritten in a compact power-law form:

$$\tau(\phi, D_f) = A(D_f) \phi^{\alpha(D_f)} \quad (4)$$

With $A(D_f) = \exp(1.3945 - 0.6601 D_f)$, $\alpha(D_f) = 3.1399 - 2.0723 D_f$

Here $\alpha(D_f)$ decreases approximately linearly with D_f and approaches -1 as $D_f \rightarrow 2$, implying that the porosity scaling of tortuosity tends toward the Euclidean-like behavior $\tau \propto \phi^{-1}$ in the Euclidean limit.

4.3 Enhancement factor relative to the Euclidean reference

To quantify how fractal embedding modifies transport relative to the Euclidean reference, we define an enhancement factor

$$E(\phi, D_f) = \frac{\tau(\phi, D_f)}{\tau_E(\phi)} \quad (5)$$

Using $\tau_E \approx 1/\phi$ as a practical Euclidean baseline, this reduces to

$$E(\phi, D_f) = \tau(\phi, D_f) \cdot \phi \quad (6)$$

Fig4.(a) shows E as a function of porosity for representative fractal dimensions. Overall, E increases with ϕ , indicating that the relative enhancement becomes more pronounced in more open pore spaces. The data also reveal a clear non-monotonic dependence on D_f : the enhancement is strongest in an intermediate range of fractal dimension (approximately $D_f \sim 1.75 - 1.80$), whereas the enhancement weakens as D_f approaches the Euclidean limit. The larger point-to-point fluctuations observed for

$D_f \approx 1.66$ and 1.73 are mainly attributed to fewer realizations (smaller n) in those cases, which increases sampling variability and leads to less stable estimates of the mean.

A global overview of $E(\varphi, D_f)$ is provided in Fig.4 (b). The heat map confirms that the maximum enhancement is concentrated along a ridge-like region in parameter space. The extracted peak ridge $D_f^*(\varphi)$ lies predominantly around $D_f \approx 1.79$ (with occasional shifts toward $D_f \approx 1.73$), demonstrating that the strongest enhancement is associated with intermediate fractal dimensions rather than the largest D_f . Over the explored dataset, E ranges from approximately 0.99 to 1.32, with the highest values occurring at high porosity and intermediate D_f . Minor deviations below unity may appear in low-porosity, low- D_f cases, which can be attributed to finite-sampling effects and the approximate nature of the Euclidean reference $\tau_E \approx 1/\varphi$.

Fig. 4 (a) Enhancement factor E versus porosity φ for representative fractal dimensions. Symbols denote numerical results from restricted random walks (error bars indicate 95% confidence intervals of the mean when available). Solid lines are weighted power-law fits of the form $E = C \varphi^\beta$ for each D_f (weights proportional to the number of realizations nnn), shown as guides to the eye.

Fig. 4(b) Heat map of $E(\varphi, D_f)$ over the (φ, D_f) parameter space. Labeled contour lines indicate round-number levels of E . The dashed curve marks the peak ridge $D_f^*(\varphi)$, i.e., the fractal dimension that maximizes E at each porosity.

5. Conclusion

Fractal phenomena are ubiquitous in nature, and uncovering the transport laws that emerge in fractal geometries remains an important and challenging topic. In this work, we investigated the tortuosity of random porous media embedded in fractal spaces constructed from an improved random Sierpiński carpet. Tortuosity was evaluated using a finite random-walk framework, and its dependence on porosity φ and fractal dimension D_f was quantified and benchmarked against the Euclidean two-dimensional case.

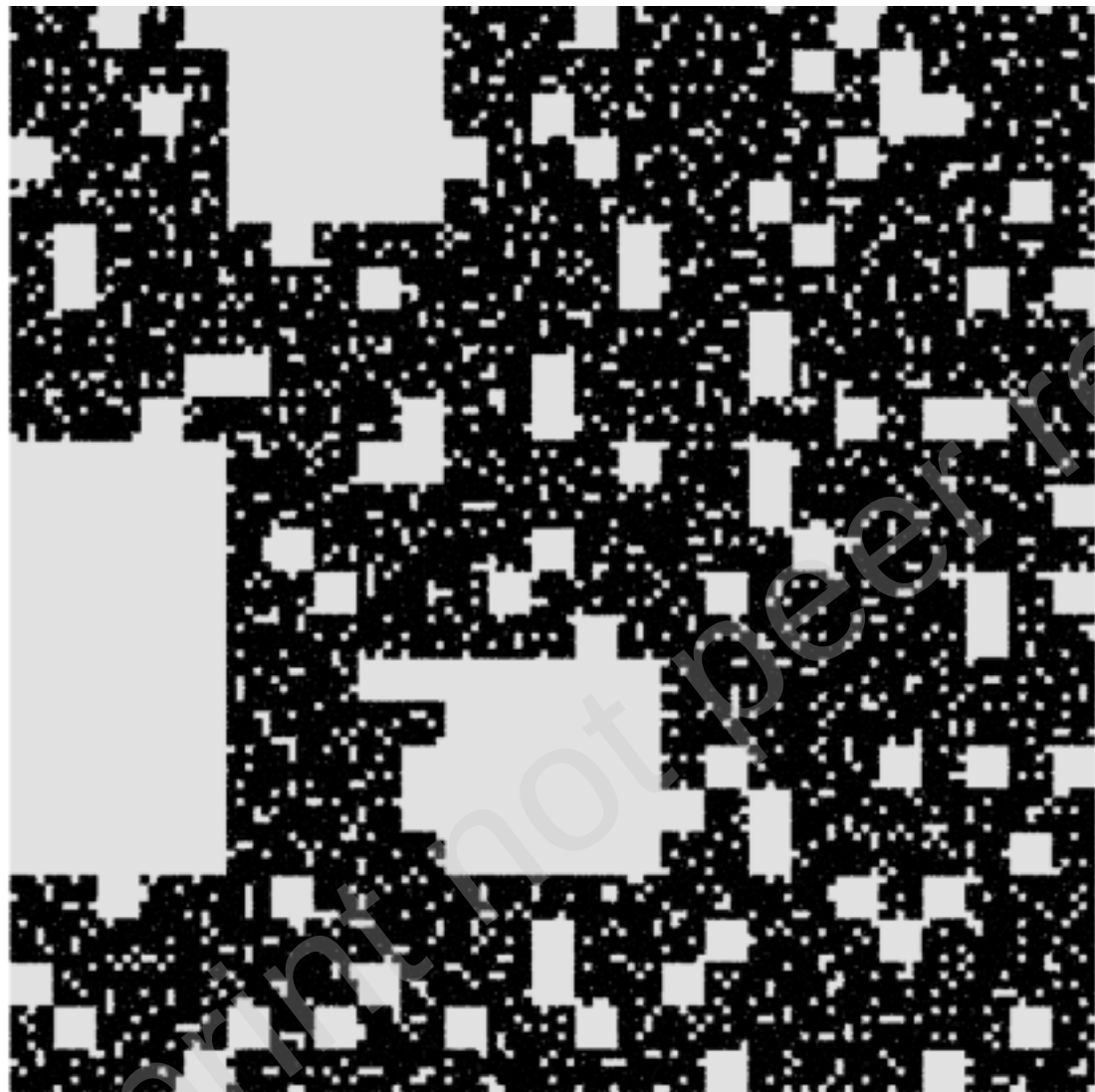
The results confirm that tortuosity generally decreases with increasing porosity in both Euclidean and fractal spaces. More importantly, the influence of fractal dimension is not purely additive: the effect of D_f depends on φ , indicating a nontrivial coupling between pore availability and fractal geometry. Compared with Euclidean media at the same porosity, random porous media in fractal spaces exhibit systematically larger tortuosity, demonstrating an intrinsic enhancement of pathway complexity induced by the fractal embedding. This discrepancy can be attributed to two complementary mechanisms. First, the random Sierpiński carpet reduces the effective space-filling capacity and thus lowers the equivalent porosity available for transport. Second, even after accounting for porosity effects, the fractal geometry introduces additional connectivity constraints (e.g., bottlenecks, dead ends, and heterogeneous channeling) that lead to transport scaling distinct from that of Euclidean space.

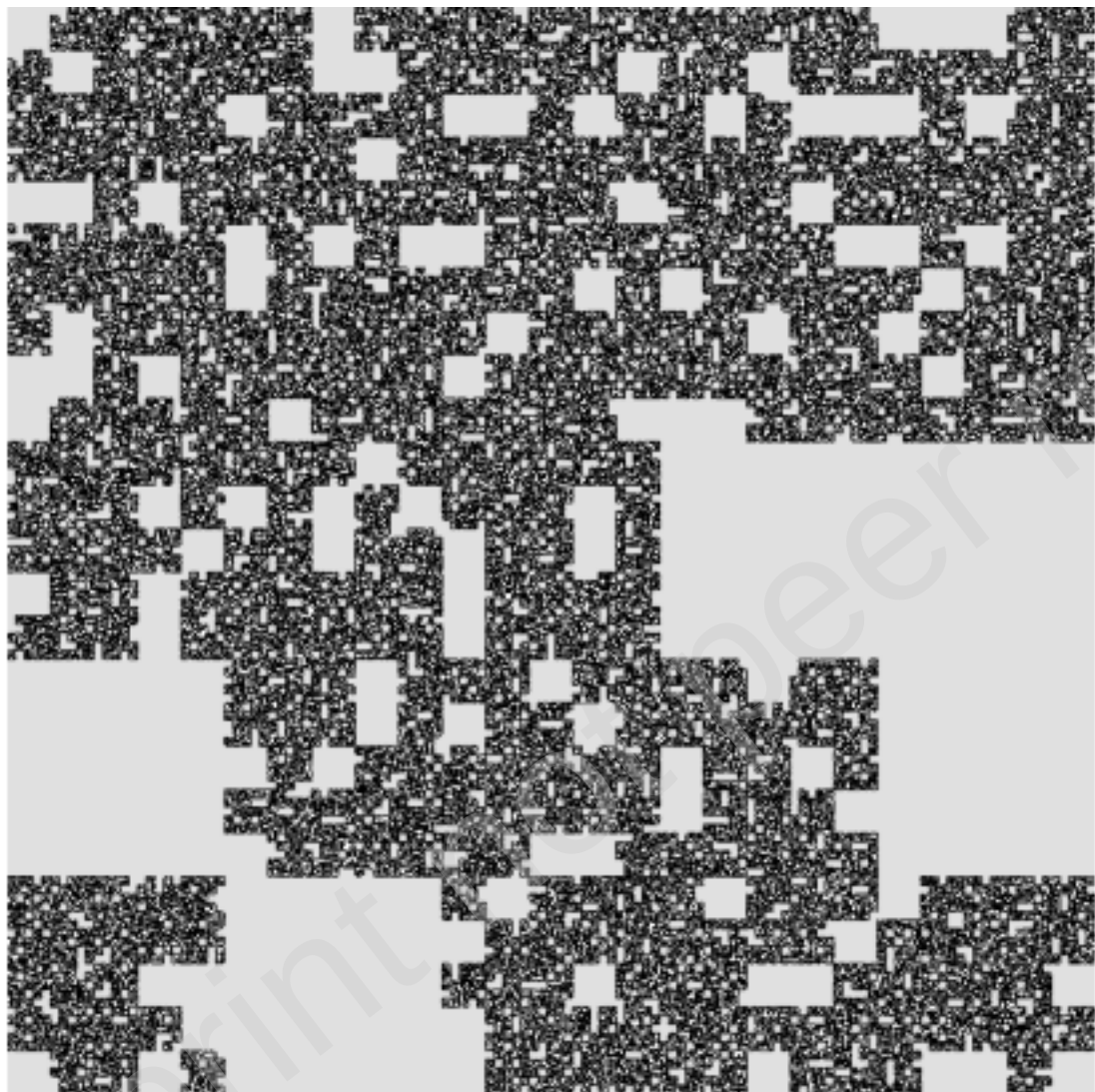
Future work will extend the present framework to other transport properties, such as permeability and thermal conductivity, and will further examine how fractal geometry controls macroscopic constitutive behavior through connectivity and scaling effects. Overall, the present study provides a quantitative basis for understanding tortuosity in fractal porous spaces and offers a reference for broader physical investigations of transport in fractal media.

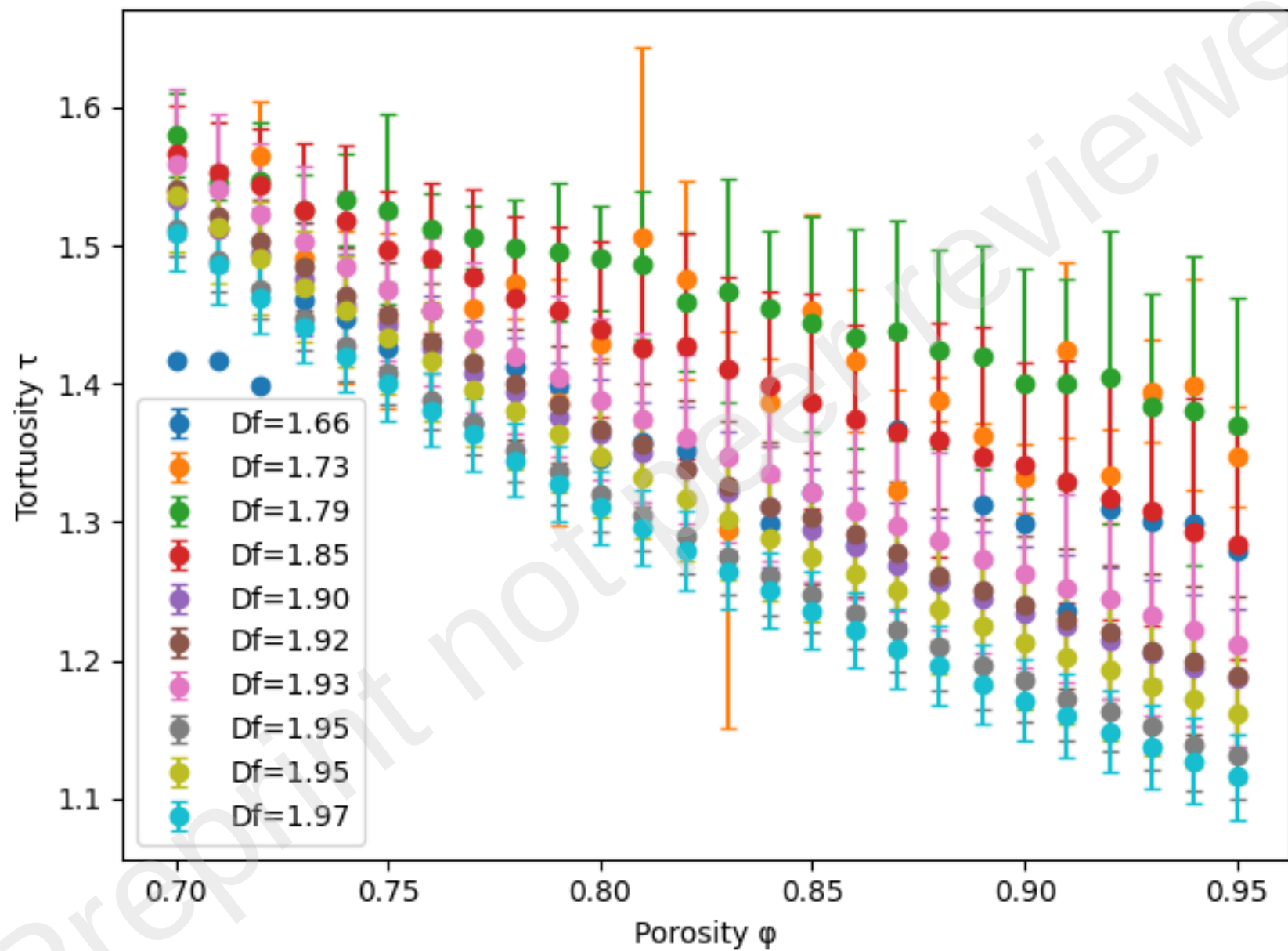
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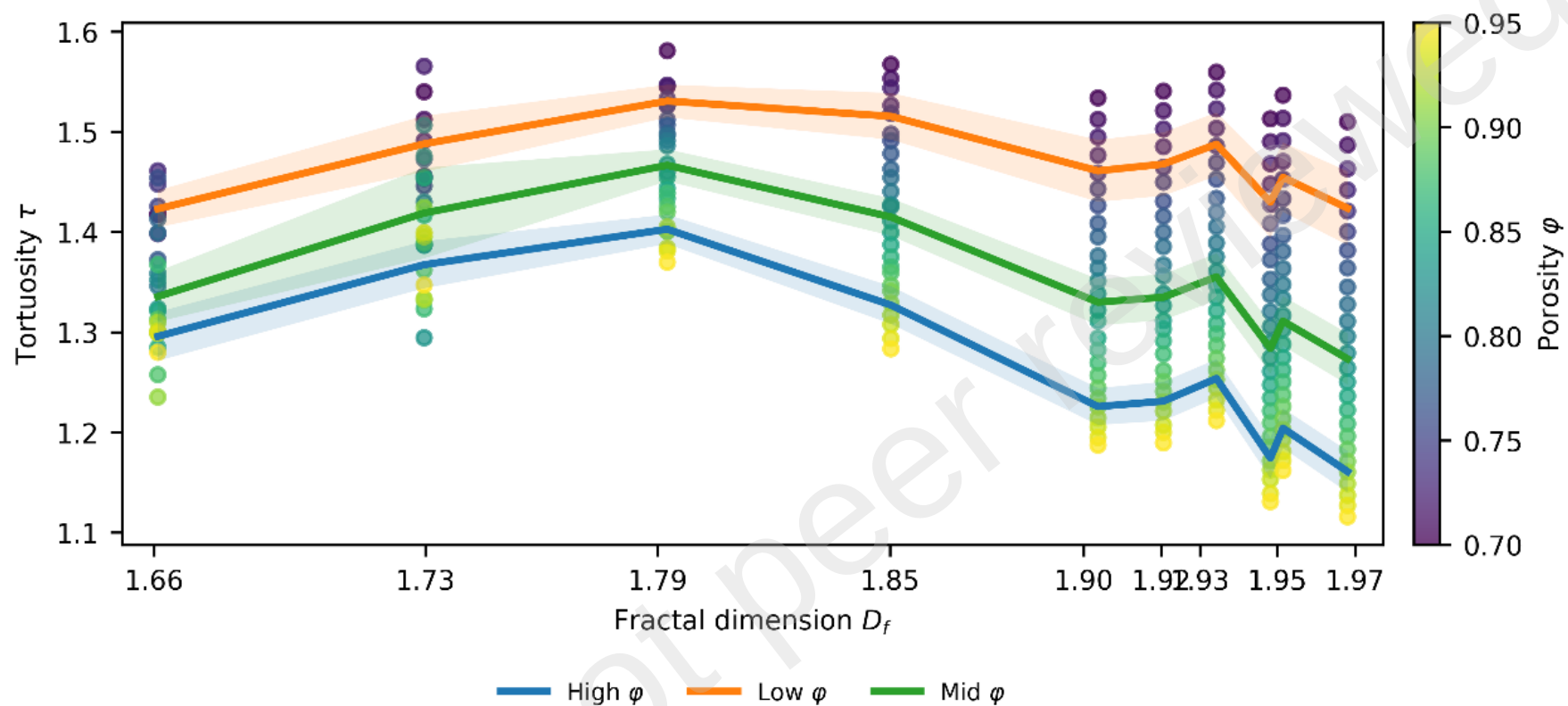
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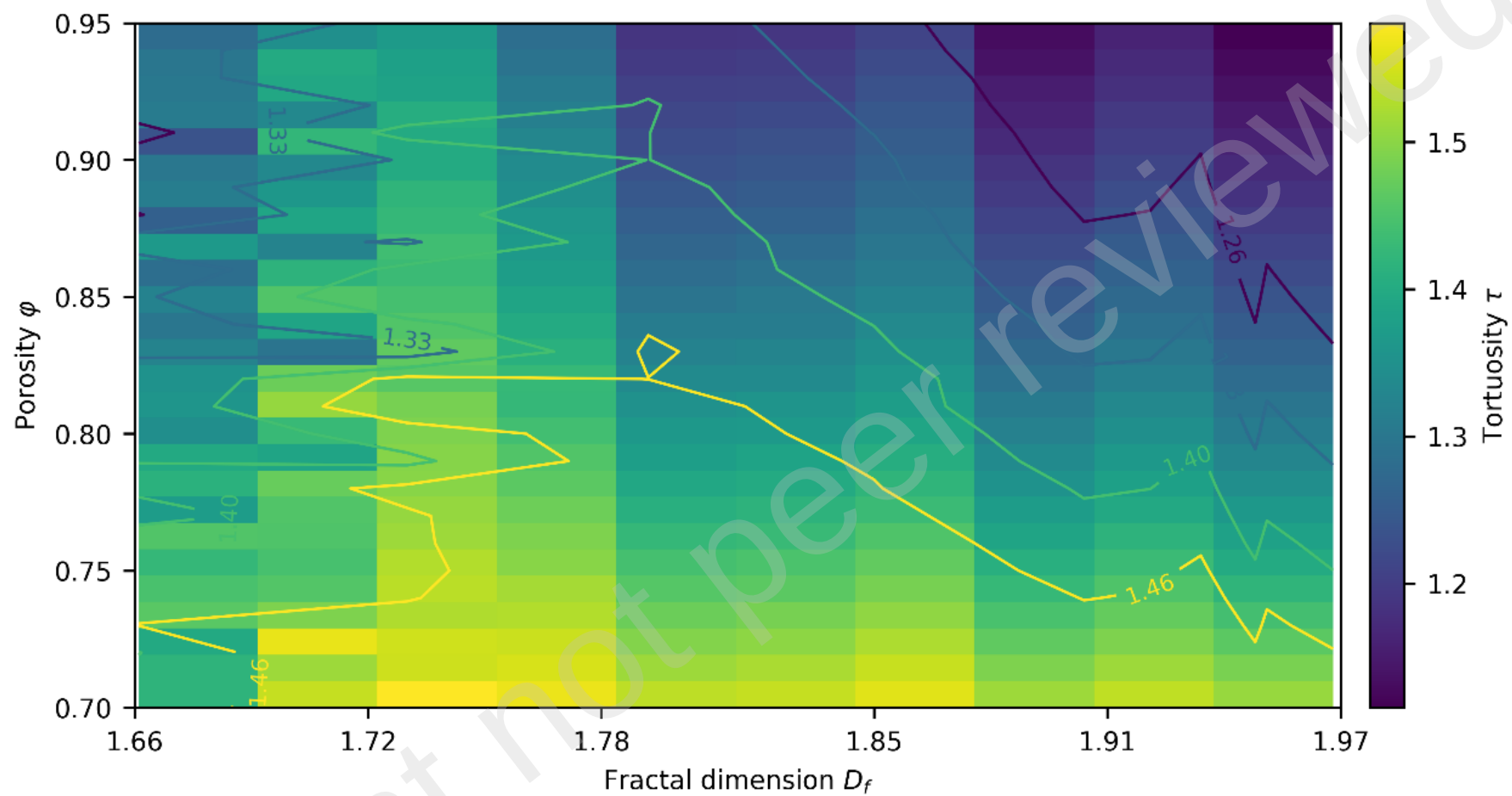
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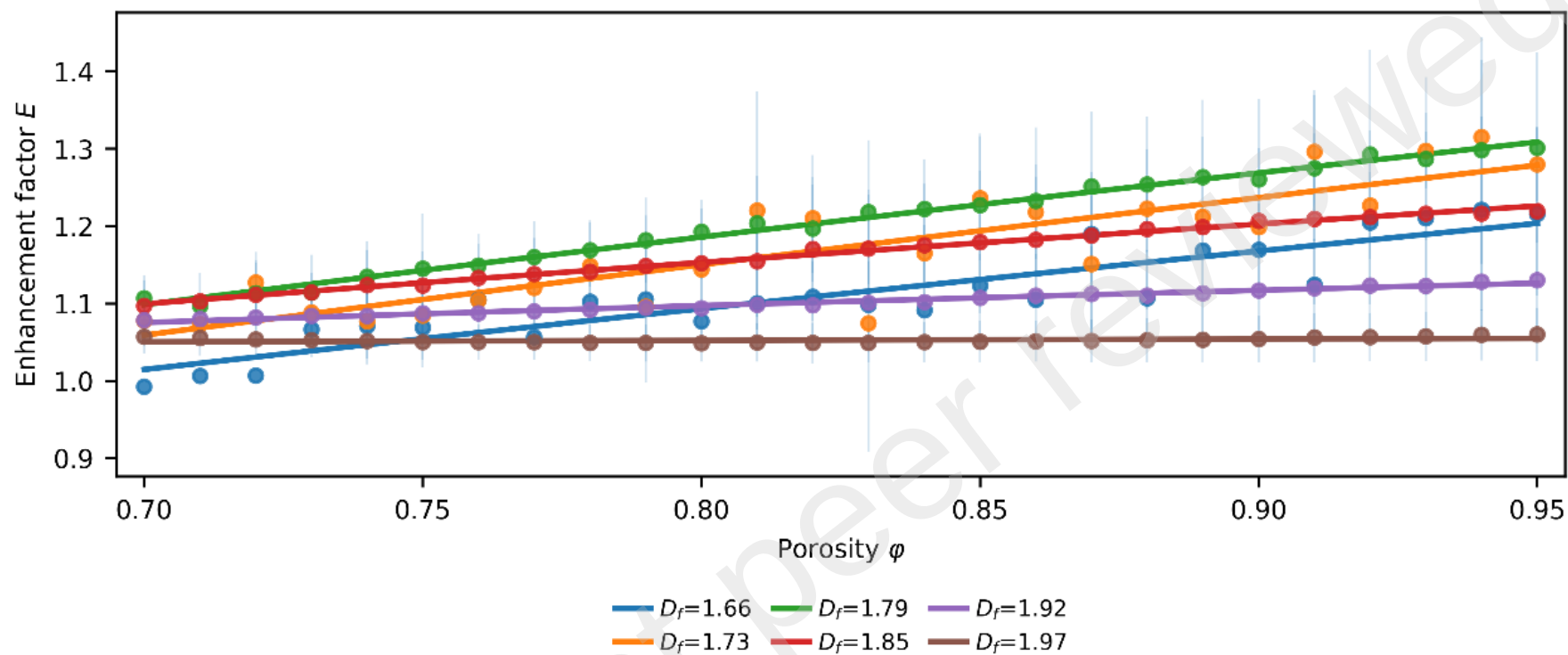


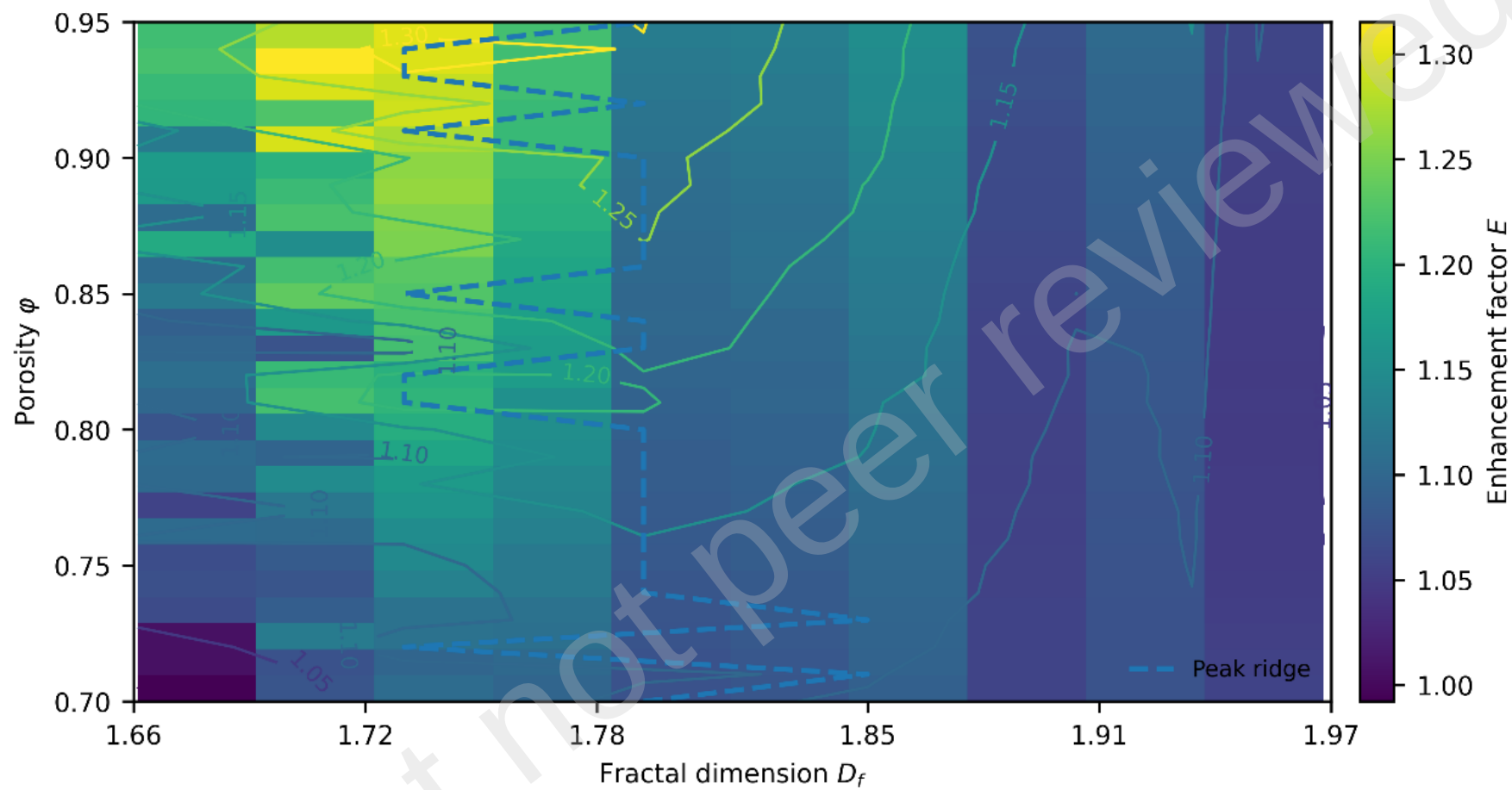












Supplementary Table 1. Mean tortuosity τ ($\pm$ standard deviation) as a function of porosity for different fractal dimensions.

(Values are rounded to two decimals. When the standard deviation is unavailable, only the mean is shown.)

Porosity ϕ	Fractal dimensions (D_f)									
	1.661	1.730	1.792	1.850	1.904	1.921	1.934	1.948	1.951	1.968
0.7	1.42	1.54	1.58 $\pm$ 0.03	1.57 $\pm$ 0.03	1.53 $\pm$ 0.03	1.54 $\pm$ 0.03	1.56 $\pm$ 0.05	1.51 $\pm$ 0.02	1.54 $\pm$ 0.04	1.51 $\pm$ 0.03
0.71	1.42	1.51	1.55 $\pm$ 0.01	1.55 $\pm$ 0.04	1.51 $\pm$ 0.03	1.52 $\pm$ 0.03	1.54 $\pm$ 0.05	1.49 $\pm$ 0.02	1.51 $\pm$ 0.04	1.49 $\pm$ 0.03
0.72	1.4	1.57 $\pm$ 0.04	1.55 $\pm$ 0.04	1.54 $\pm$ 0.04	1.49 $\pm$ 0.03	1.50 $\pm$ 0.03	1.52 $\pm$ 0.05	1.47 $\pm$ 0.02	1.49 $\pm$ 0.04	1.46 $\pm$ 0.03
0.73	1.46	1.49	1.53 $\pm$ 0.03	1.53 $\pm$ 0.05	1.48 $\pm$ 0.04	1.49 $\pm$ 0.03	1.50 $\pm$ 0.05	1.45 $\pm$ 0.02	1.47 $\pm$ 0.04	1.44 $\pm$ 0.03
0.74	1.45	1.46 $\pm$ 0.05	1.53 $\pm$ 0.03	1.52 $\pm$ 0.05	1.46 $\pm$ 0.04	1.46 $\pm$ 0.03	1.49 $\pm$ 0.05	1.43 $\pm$ 0.03	1.45 $\pm$ 0.04	1.42 $\pm$ 0.03
0.75	1.43	1.45 $\pm$ 0.06	1.53 $\pm$ 0.07	1.50 $\pm$ 0.04	1.44 $\pm$ 0.03	1.45 $\pm$ 0.04	1.47 $\pm$ 0.05	1.41 $\pm$ 0.02	1.43 $\pm$ 0.04	1.40 $\pm$ 0.03
0.76	1.45	1.45	1.51 $\pm$ 0.03	1.49 $\pm$ 0.05	1.43 $\pm$ 0.04	1.43 $\pm$ 0.04	1.45 $\pm$ 0.05	1.39 $\pm$ 0.02	1.42 $\pm$ 0.04	1.38 $\pm$ 0.03
0.77	1.37	1.45	1.51 $\pm$ 0.02	1.48 $\pm$ 0.06	1.41 $\pm$ 0.04	1.42 $\pm$ 0.04	1.43 $\pm$ 0.05	1.37 $\pm$ 0.02	1.40 $\pm$ 0.04	1.36 $\pm$ 0.03
0.78	1.41	1.47 $\pm$ 0.03	1.50 $\pm$ 0.04	1.46 $\pm$ 0.06	1.40 $\pm$ 0.04	1.40 $\pm$ 0.04	1.42 $\pm$ 0.06	1.35 $\pm$ 0.02	1.38 $\pm$ 0.04	1.34 $\pm$ 0.03
0.79	1.4	1.39 $\pm$ 0.09	1.50 $\pm$ 0.05	1.45 $\pm$ 0.06	1.38 $\pm$ 0.04	1.39 $\pm$ 0.04	1.41 $\pm$ 0.06	1.34 $\pm$ 0.03	1.36 $\pm$ 0.04	1.33 $\pm$ 0.03
0.8	1.35	1.43 $\pm$ 0.01	1.49 $\pm$ 0.04	1.44 $\pm$ 0.06	1.36 $\pm$ 0.04	1.37 $\pm$ 0.05	1.39 $\pm$ 0.06	1.32 $\pm$ 0.03	1.35 $\pm$ 0.04	1.31 $\pm$ 0.03
0.81	1.36	1.51 $\pm$ 0.14	1.49 $\pm$ 0.05	1.43 $\pm$ 0.06	1.35 $\pm$ 0.04	1.36 $\pm$ 0.04	1.38 $\pm$ 0.06	1.31 $\pm$ 0.03	1.33 $\pm$ 0.04	1.30 $\pm$ 0.03
0.82	1.35	1.48 $\pm$ 0.07	1.46 $\pm$ 0.05	1.43 $\pm$ 0.08	1.34 $\pm$ 0.05	1.34 $\pm$ 0.05	1.36 $\pm$ 0.06	1.29 $\pm$ 0.03	1.32 $\pm$ 0.05	1.28 $\pm$ 0.03
0.83	1.32	1.29 $\pm$ 0.14	1.47 $\pm$ 0.08	1.41 $\pm$ 0.07	1.32 $\pm$ 0.04	1.33 $\pm$ 0.05	1.35 $\pm$ 0.06	1.27 $\pm$ 0.03	1.30 $\pm$ 0.04	1.26 $\pm$ 0.03
0.84	1.3	1.39 $\pm$ 0.03	1.45 $\pm$ 0.06	1.40 $\pm$ 0.07	1.31 $\pm$ 0.04	1.31 $\pm$ 0.05	1.33 $\pm$ 0.06	1.26 $\pm$ 0.03	1.29 $\pm$ 0.04	1.25 $\pm$ 0.03
0.85	1.32	1.45 $\pm$ 0.07	1.44 $\pm$ 0.08	1.39 $\pm$ 0.08	1.29 $\pm$ 0.04	1.30 $\pm$ 0.05	1.32 $\pm$ 0.06	1.25 $\pm$ 0.03	1.27 $\pm$ 0.05	1.24 $\pm$ 0.03
0.86	1.28	1.42 $\pm$ 0.05	1.43 $\pm$ 0.08	1.37 $\pm$ 0.07	1.28 $\pm$ 0.04	1.29 $\pm$ 0.05	1.31 $\pm$ 0.06	1.23 $\pm$ 0.03	1.26 $\pm$ 0.05	1.22 $\pm$ 0.03
0.87	1.37	1.32 $\pm$ 0.07	1.44 $\pm$ 0.08	1.37 $\pm$ 0.07	1.27 $\pm$ 0.04	1.28 $\pm$ 0.05	1.30 $\pm$ 0.06	1.22 $\pm$ 0.03	1.25 $\pm$ 0.05	1.21 $\pm$ 0.03
0.88	1.26	1.39 $\pm$ 0.02	1.43 $\pm$ 0.07	1.36 $\pm$ 0.08	1.26 $\pm$ 0.05	1.26 $\pm$ 0.05	1.29 $\pm$ 0.06	1.21 $\pm$ 0.03	1.24 $\pm$ 0.05	1.20 $\pm$ 0.03
0.89	1.31	1.36 $\pm$ 0.01	1.42 $\pm$ 0.08	1.35 $\pm$ 0.09	1.24 $\pm$ 0.05	1.25 $\pm$ 0.05	1.27 $\pm$ 0.07	1.20 $\pm$ 0.03	1.23 $\pm$ 0.05	1.18 $\pm$ 0.03
0.9	1.3	1.33 $\pm$ 0.02	1.40 $\pm$ 0.08	1.34 $\pm$ 0.08	1.23 $\pm$ 0.05	1.24 $\pm$ 0.05	1.26 $\pm$ 0.07	1.19 $\pm$ 0.03	1.21 $\pm$ 0.05	1.17 $\pm$ 0.03
0.91	1.24	1.42 $\pm$ 0.06	1.40 $\pm$ 0.08	1.33 $\pm$ 0.09	1.23 $\pm$ 0.05	1.23 $\pm$ 0.05	1.25 $\pm$ 0.07	1.17 $\pm$ 0.03	1.20 $\pm$ 0.05	1.16 $\pm$ 0.03
0.92	1.31	1.33 $\pm$ 0.03	1.40 $\pm$ 0.11	1.32 $\pm$ 0.09	1.21 $\pm$ 0.05	1.22 $\pm$ 0.05	1.24 $\pm$ 0.07	1.16 $\pm$ 0.03	1.19 $\pm$ 0.05	1.15 $\pm$ 0.03
0.93	1.3	1.40 $\pm$ 0.04	1.38 $\pm$ 0.08	1.31 $\pm$ 0.08	1.21 $\pm$ 0.05	1.21 $\pm$ 0.06	1.23 $\pm$ 0.07	1.15 $\pm$ 0.03	1.18 $\pm$ 0.05	1.14 $\pm$ 0.03
0.94	1.3	1.40 $\pm$ 0.08	1.38 $\pm$ 0.11	1.29 $\pm$ 0.10	1.19 $\pm$ 0.05	1.20 $\pm$ 0.05	1.22 $\pm$ 0.07	1.14 $\pm$ 0.03	1.17 $\pm$ 0.05	1.13 $\pm$ 0.03
0.95	1.28	1.35 $\pm$ 0.04	1.37 $\pm$ 0.09	1.28 $\pm$ 0.08	1.19 $\pm$ 0.05	1.19 $\pm$ 0.06	1.21 $\pm$ 0.07	1.13 $\pm$ 0.03	1.16 $\pm$ 0.05	1.12 $\pm$ 0.03